GEOG 4/5/7 9073: Environmental Analysis in R

Week 7.01: More spatial data operations

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Today's schedule

- Open discussion
- Schedule "stuff"
- Topology and the "9-intersection model"
- Spatial data operations
- Activity

Anything to discuss? Questions?

Updates

- **This week** will cover "weeks 6 & 7" on Canvas
- **Next week** will catch-up by completing week 8 (point-pattern analysis)
- Then...
    - (wk9) Spring Break
    - (wk10) "Off" again for AAG (big geography conference)
    - (wk11) Will meet Tuesday (March 24th), off Thursday (March 26th)

today's setup

```
library(tidyverse)
library(GISTools)
library(sf)
library(tmap)
```

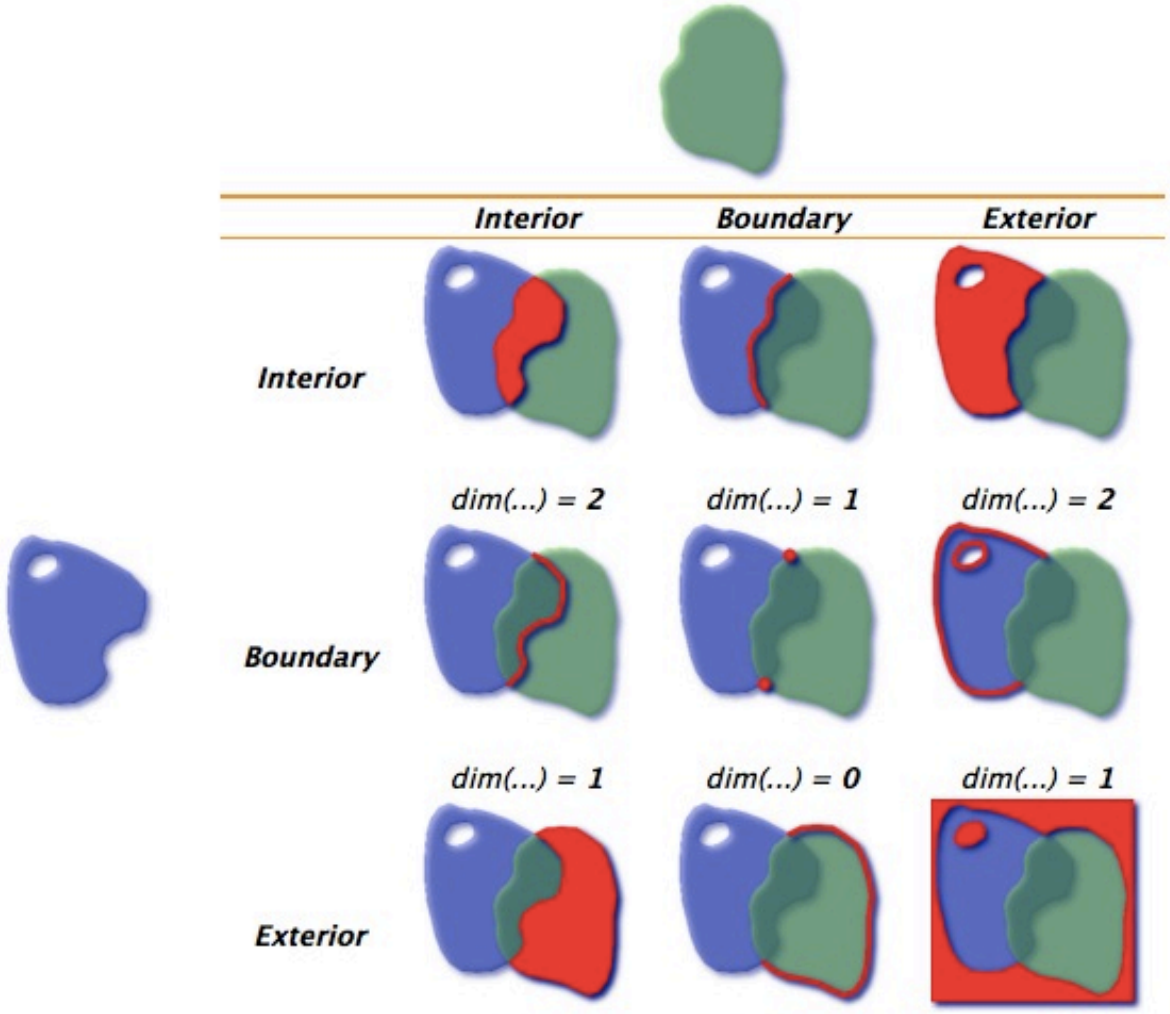
And follow along with `./src/week07_inclass.Rmd` from the class repo

What does "topology" mean?

...and "ontology"?

and why should we care?

The "9-intersection model"



Spatial predicates

- Equals
- Disjoint
- Touches (meets)
- Contains
- Covers
- Intersects
- Within (inside)
- Covered by
- Crosses
- Overlaps

Spatial predicates in `sf::`

- Equals `st_equals`
- Disjoint `st_disjoint`
- Touches (meets) `st_touches`
- Contains `st_contains`
- Covers `st_covers`
- Intersects `st_intersects`
- Within (inside) `st_within`
- Covered by `st_covered_by`
- Crosses `st_crosses`
- Overlaps `st_overlaps`

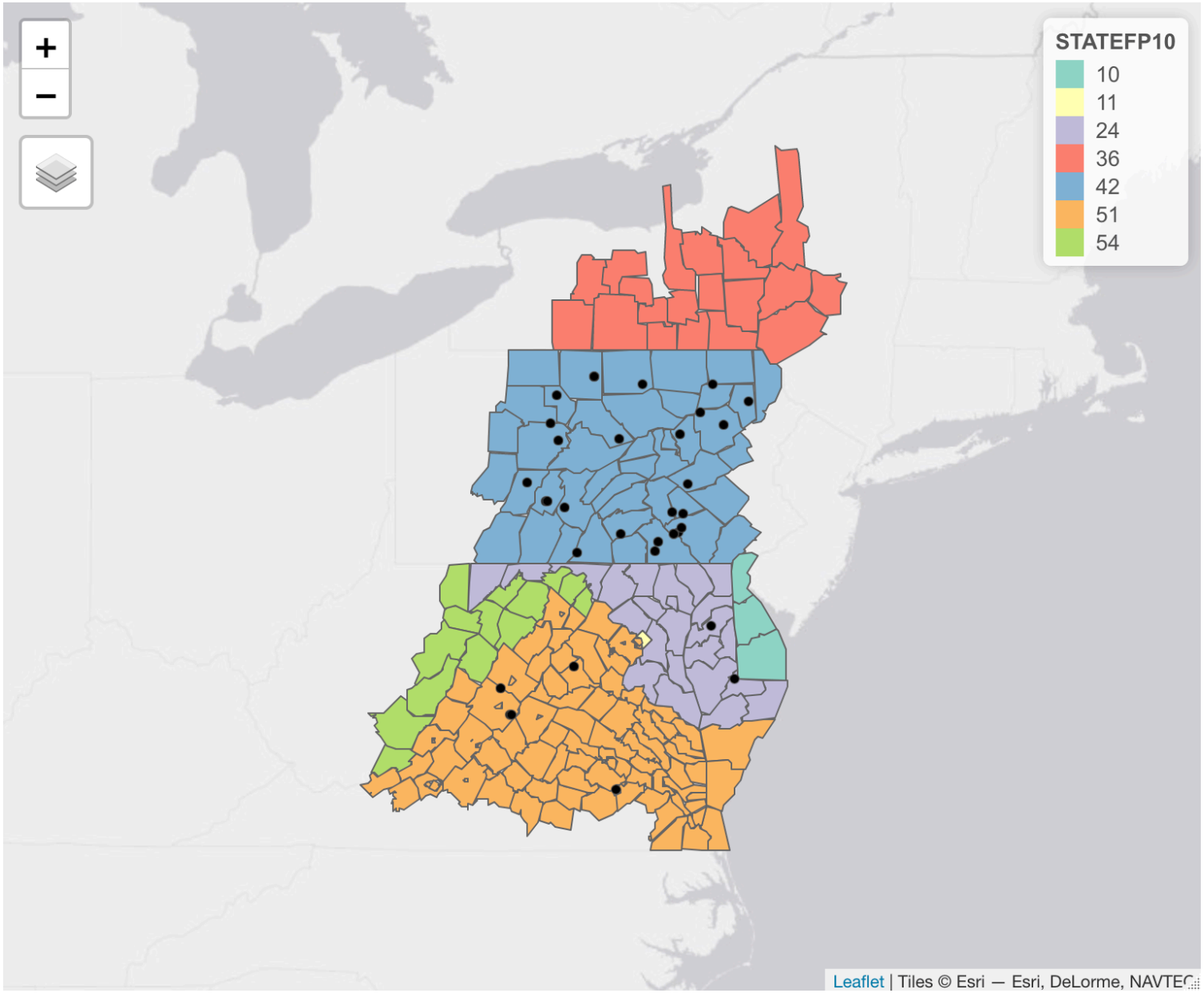
Let's try some of them

Today's data

```r
counties <- sf::read_sf("./data/CBW/County_Boundaries.shp") %>% sf::st_make_valid()

dams <- sf::read_sf("./data/CBW/Dam_or_Other_Blockage_Removed_2012_2017.shp") %>% sf::st_make_valid()

streams <- sf::read_sf("./data/CBW/Streams_Opened_by_Dam_Removal_2012_2017.shp") %>% sf::st_make_valid()
```


STATEFP10
10
11
24
36
42
51
54
Leaflet | Tiles © Esri — Esri, DeLorme, NAVTEC

Remind me, how would we find the dams that are in Pennsylvania?

First, let's get the counties in PA

```
pa.counties <- counties %>% filter(STATEFP10 == 42)
```

Dams in PA

Multiple methods

```
pa.dams <- st_intersection(dams, pa.counties)

# or

pa.dams <- dams[pa.counties,]
```

How do these functions work? (and what do they output?)

Let's use a predicate to see the Boolean result

```
st_intersects(dams, pa.counties)
```

What's the output?

Order matters

```
# order matters
dams %>% st_intersects(x = ., y = pa.counties)
dams %>% st_intersects(x = pa.counties, y = .)
```

Matrix output with the `sparse` parameter

Try this

```
# get a dense logical matrix
dams %>% st_intersects(x = ., y = pa.counties, sparse = F)
```

How should we interpret the output?

Other predicates

```
# Disjoint
dams %>% st_disjoint(., pa.counties, sparse = F)

# Within
dams %>% st_within(., pa.counties, sparse = F)
```

More examples

```r
c.tioga <- counties %>%
  dplyr::filter(str_detect(NAME10, "Tioga") & STATEFP10 == 42)

streams.tioga <- streams %>% st_intersection(., c.tioga)

# double check the work
streams.tioga %>% st_covered_by(., c.tioga)
tm_shape(c.tioga) + tm_polygons(fill = "tan") + tm_shape(streams.tioga) + tm_lines(col = "blue")
```

distance

```r
streams.tioga %>% st_is_within_distance(., dams, 1)
```

But wait, what's the units?

Spatial joins

How do *tabular* joins work?

Spatial joins are similar, but use spatial relationships

How we normally think about spatial joins...

if A intersects B, then take the attributes from B, and join them to A
(and discard all elements from B where "intersects" is false)

- This is how ArcGIS Pro works
- But we don't need to restrict ourselves to the "intersects" predicate

calling `sf::st_join`

From the help menu...

```
st_join(
  x,
  y,
  join = st_intersects,
  ...,
  suffix = c(".x", ".y"),
  left = TRUE,
  largest = FALSE
)
```

A spatial join the "typical" way

```
st_join(pa.counties, dams, join = st_intersects)
```

But we can perform spatial joins using other predicates...

```
st_join(pa.counties, dams, join = st_disjoint)
```

or...

```
st_join(pa.counties, streams, join = st_touches)
```

Think back to the 9-intersection model…

and explore (try) some more yourself!

Part 2

A story from my Vermont research

- Nutrient runoff (primarily phosphorus) causes harmful cyanobacteria blooms in Lake Champlain
- TMDL regulations "limit" the amount of criteria pollutants allowed to enter water bodies that flow to the Lake
- To regulate, and to improve, we need to measure
- So EPA has broken-down the Lake Champlain Basin into units of analysis called **NHDPlus segments**

But management is a different issue

- We don't commonly manage water-related problems using watershed-derived jurisdictions
- We should, but we don't
- In Vermont, they're doing something a bit different
- Tactical Basins, CWSPs, and RPCs

Let's estimate the load in each RPC

Start with the NHDs

```r
# NHDs
nhds <- sf::read_sf("./data/nhdplus_loads.shp") %>% sf::st_make_valid()
glimpse(nhds)
tm_shape(nhds) + tm_polygons(fill = "Baseline_L", n = 10)
```

`Baseline_L` is phosphorus load in kg/year

And now the RPCs

```
# RPCs
rpcs <- sf::read_sf("./data/gn_vt_rpcs.shp") %>% sf::st_make_valid()
glimpse(rpcs)
tm_shape(rpcs) + tm_polygons(fill = "INITIALS")
```

Overlay them

```
tm_shape(rpcs) +
  tm_shape(nhds) + tm_polygons(fill = "Baseline_L", n = 7) +
  tm_shape(rpcs) + tm_borders(col = "red")
```

Why am I calling `tm_shape(rpcs)` twice? Any ideas?

Our objective:

Calculate the total phosphorus load in each RPC

Ideas for how to accomplish that task?

A spatial join

```r
# do the join
nhd_rpcs <- st_join(nhds, rpcs, join = st_intersects)

# look at it/confirm it worked
glimpse(nhd_rpcs)

# plot it
tm_shape(nhd_rpcs) + tm_polygons(fill = "RPC")
```

and now to summarize...

Suggestions?

The "tidy way"

```
# the "tidy way"
nhd_rpcs %>%
  group_by(RPC) %>%
  summarize(totalLoad = sum(Baseline_L))
```

What was the output?

Add a line to plot it

```
nhd_rpcs %>%
  group_by(RPC) %>%
  summarize(totalLoad = sum(Baseline_L)) %>%
  tm_shape(.) + tm_polygons(fill = "totalLoad") # <- this line is new
```

the utility of pipes

An alternative method

from your book...

- works a bit differently
- uses `sf` features to do the aggregation

```
# using aggregate instead
aggregate(x = nhds, by = rpcs, FUN = sum) # throws an error... what's the problem?
```

Error message:

```
Error in FUN(X[[i]], ...) : invalid 'type' (character) of argument
```

What's the problem?

Fixing the problem

```
glimpse(nhds) # How many character attributes?

# fix the problem
nhds %>% dplyr::select(-SOURCEFC, -NHDPlus_Ca, -Tactical_B) %>%
  aggregate(x = ., by = rpcs, FUN = sum)
```

Do it again, but assign it to a variable

```
# same function, but assign it to a variable
agg.rpcs <- nhds %>% dplyr::select(-SOURCEFC, -NHDPlus_Ca, -Tactical_B) %>%
  aggregate(x = ., by = rpcs, FUN = sum)
```

Plot it

```
# plot it... why and how is it different?
tm_shape(agg.rpcs) + tm_polygons(fill = "Baseline_L")
```

how/why is it different than our earlier plot?

So, have we completed our task?

Any remaining issues?

Let's check our spatial relationships

Break down the code

```
# issues with overlap
nhd_rpcs %>% group_by(NHDPlus_ID) %>% summarise(count = n()) %>%
  arrange(desc(count))
```

What's the result?

Check with a quick plot

```
# same as above, but plot it
nhd_rpcs %>% group_by(NHDPlus_ID) %>% summarise(count = n()) %>%
  arrange(desc(count)) %>%
  mutate(count = as.character(count)) %>% # this is a shortcut,
  # and while it works, not best practice
  tm_shape(.) + tm_polygons(fill = "count")
```

Area-weighted interpolation

(to the whiteboard!)

```r
# area-weighted interpolation
interp.loads <- nhds %>% dplyr::select(Baseline_L, geometry) %>%
  st_interpolate_aw(., rpcs, extensive = T)

tm_shape(interp.loads) + tm_polygons(fill = "Baseline_L")
```

Same? Different? How can we compare?

Do a join

(note, I'm cheating a bit here and relying on the RPC geometry being identical)

```
# do a join
comparison <- st_join(agg.rpcs, interp.loads, st_equals)

# calculate the error, then map it
tmap_mode("view")

comparison %>% mutate(diff = Baseline_L.x - Baseline_L.y) %>%
  tm_shape(.) + tm_polygons(col = "diff") +
  tm_shape(nhds) + tm_borders(col = "blue")
```

How different were the methods?

If there's time,

return to last week's Wild Card activity...

For this week

- Thursday will be a "wild card" day, and WILL REQUIRE YOU TO HAVE READ CHAPTER 5
- Chapter 5 from Lovelace (https://geocompr.robinlovelace.net/geometric-operations.html)
- Practice, practice, practice
- Work on your projects (in-class updates after spring break)